WEB TECHNOLOGIES LAB – QUESTION PAPER

Time: 3 Hours

List of Experiments

HTML & CSS (WEEK-1 & WEEK-2)

1. Design a webpage for an online bookstore using HTML frames with navigation, content, and footer sections.
2. Create a frameset layout and link multiple pages to display book details dynamically.
3. Apply CSS styling to the bookstore webpage to improve fonts, colors, and layout.
4. Design a responsive webpage using CSS with proper alignment, margins, and padding.

JavaScript (WEEK-3 & WEEK-4)

5. Develop a user registration form with validation for username, password, and confirm password.
6. Create a forgot password form with email validation and appropriate error messages.
7. Implement dynamic content loading using JavaScript when clicking book links.
8. Design a simple shopping cart that allows users to add and view selected books.
9. Enhance the shopping cart by adding remove and total price calculation features.
10. Create a form validation system using JavaScript for multiple input fields.

XML (WEEK-5 & WEEK-6)

11. Design an XML structure to represent a book catalog with title, author, genre, price, and availability.
12. Create an XML file to store user account details including username, password, email, and purchase history.
13. Validate an XML document using DTD or XML Schema.
14. Display XML data in a webpage using JavaScript or XSLT.

Server Setup & Servlets (WEEK-7 & WEEK-8)

15. Install and configure Apache Tomcat server in Eclipse IDE.
16. Develop a servlet to display book details retrieved from a database.
17. Create a servlet that handles user input and displays dynamic content.

JSP & Authentication (WEEK-9)

18. Develop a login page using JSP and implement user authentication.
19. Restrict access to certain pages based on user login status.

Hibernate (WEEK-10)

20. Create Hibernate entity classes for books and users and map them to database tables.
21. Perform CRUD operations using Hibernate for book records.

Advanced HTML, CSS & JavaScript

22. Design a webpage using HTML5 semantic tags (header, section, article, footer) for an online bookstore.
23. Create a CSS-based navigation menu with hover effects and dropdown options.
24. Implement a JavaScript-based image slider to display featured books.
25. Develop a form with real-time validation using JavaScript (without submit button).

Advanced JavaScript & Shopping Cart

26. Enhance the shopping cart to store data using browser Local Storage.
27. Implement a feature to update the quantity of books in the shopping cart dynamically.

XML & Data Handling

28. Create an XML Schema (XSD) for validating the book catalog XML structure.
29. Convert XML data into a formatted HTML page using XSLT.

Servlets & JSP

30. Develop a servlet program to handle session management (login/logout functionality).
31. Create a JSP page to display database records using JDBC connectivity.